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Deep Neural Network for Functional Graphical Models Structure Learning

Shuoyang Wang[a] and Guanqun Cao[b,c]

[a]Department of Bioinformatics and Biostatistics, University of Louisville, Louisville, KY; [b]Department of Statistics and Probability, Michigan State University, East Lansing, MI; [c]Department of Computational Mathematics, Science and Engineering, Michigan State University, East Lansing, MI

ABSTRACT

Functional data refer to data that are realizations of random functions varying over a continuum, such as images or signals. In many modern fields, including neuroscience, medical science, and traffic monitoring, observations are better modeled as multivariate random functions rather than as vectors. To capture the conditional independence structure of such multivariate functional data, functional graphical models have been developed. In this article, we propose a novel and flexible method to estimate the neighborhood of each node using a deep neural network-based functional data regression and feature selection approach with an arbitrary nonparametric form. The full graph structure is then recovered by combining the estimated neighborhoods. Our approach avoids common distributional assumptions on the random functions and circumvents the need for a well-defined precision operator, which may not exist in the functional data context. Furthermore, we establish model consistency for the proposed algorithm. The convergence rate reaches to the classical nonparametric regression rate up to a logarithmic factor. We discover a novel critical sampling frequency that governs the convergence rates of the deep neural network estimator for both densely and sparsely observed functional data. The empirical performance of our method is demonstrated through simulation studies and a real data application. Supplementary materials for this article are available online.

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Deep learning; Functional data analysis; Functional graphical model; Feature selection; Neighborhood selection

1. Introduction

Graphical models are powerful tools for describing the conditional independence structures of random vector-valued data. However, many applications involve functional data, which consist of realizations of random functions that vary over a continuum, such as a time interval or a spatial domain. Common examples of functional data include signals and many emerging high-throughput digital measurements. Understanding the dependence structure of functional data is crucial in a wide range of applications. For instance, in neuroimaging data analysis, we focus on electroencephalogram (EEG) and functional magnetic resonance imaging (fMRI) data, where each sampling unit is modeled as a realization of a a realization of a p-dimensional stochastic process. The primary objective is to depict the conditional dependence structure among these p random functions.

1.1. Motivation

Our study is first motivated by a brain activity analysis (see details in Section 6). In a typical brain activity analysis in neuroscience, nodes represent brain regions and edges indicate functional connectivity over time. By analyzing fMRI data (a type of functional data), researchers use functional graphical models (FGMs) to understand how different parts of the brain communicate during various tasks or conditions. Recent research has highlighted a hierarchical organization of brain connectivity. Brain magnetic resonance images are preprocessed and segmented into temporal signals for 116 regions of interest (ROIs) through the automatic anatomical labeling parcellation approach (Tzourio-Mazoyer et al. 2002). By learning the conditional independence graph of these ROIs, we uncover key insights into these brain connectivity patterns. Furthermore, comparing conditional independence graphs between populations with and without specific neurodevelopmental conditions (e.g., autism or ADHD) could provide critical clues about the neural underpinnings of certain symptoms or disorders. The FGMs offer distinct advantages over traditional, nonfunctional analyses of fMRI data. In particular, our approach captures spatio-temporal dependencies among ROIs. For example, it can identify how neural activity in one brain region at time t_1 influences activity in another region at a later time t_2. This capability allows for a deeper understanding of the dynamic interactions within the brain.

However, most existing FGMs in the literature assume that the random functions follow a multivariate Gaussian process. While this assumption simplifies modeling, it may not accurately reflect many real-world functional data applications, which often involve continuous but non-Gaussian data. For example, EEG data recording voltage variations over time through scalp electrodes-can be significantly influenced by external

CONTACT Guanqun Cao caoguanq@msu.edu Department of Statistics and Probability, and Department of Computational Mathematics, Science and Engineering, Michigan State University, 619 Red Cedar Road Wells, Hall, Room C413, East Lansing, MI 48824.
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stimuli and muscle movements, leading to non-Gaussian characteristics. Similarly, biological and physical processes such as bacterial growth, particle sizes, and transport within biological tissues frequently exhibit nonnormal distributions. Relying on normality assumptions for such functional data can hinder the detection of important conditional dependencies, limiting the broader applicability and effectiveness of these models.

1.2. Contribution

Our goal in this work is to depict graph structure without restrict the underlying data-generating process follows Gaussian. To do so, we use deep neural networks (DNNs) framework. DNNs are estimation methods that model the relationship between inputs and outputs using layers of connected computational units. They can capture complex nonlinear dependencies, making them more versatile for real-world applications. Applying DNNs into functional data domain is emerging in the recent years. However, DNNs methods have received limited attention in the studies of FGM. We aim to investigate how to leverage DNN in learning FGM structure. We propose a new nonparametric FGM that allows the conditional relationships among the principal scores to take a fully nonparametric structure.

Specifically, we conduct the neighborhood selection procedure as a functional DNN regression problem. Due to the infinite-dimensional nature of functional data, we first extract projection scores from all observed random functions. We then approximate the functional regression as a vector-on-vector regression problem, which is solved using a DNN with a Lasso penalty. For calibration purposes, when estimating the neighborhood of a target node, we project all functions onto the same basis instead of projecting each function onto its own subspace. Furthermore, to enhance the selection of tuning parameters in the proposed functional DNN Lasso algorithm, we extend the high-dimensional BIC criterion for DNN Lasso algorithms.

In summary, the proposed DNN learning approach offers five key benefits. First, it enables us to define FGMs directly from the conditional distribution without the need to introduce a precision operator or assume that the process distribution is Gaussian or copula Gaussian. Second, while most existing neighborhood selection methods rely on either linear function-on-function regression models Zhao et al. (2024) or additive models Solea and Dette (2022), our DNN framework eliminates these parametric and semi-parametric constraints. This approach increases flexibility by allowing fully nonparametric functional relationships among different neighborhood nodes. Third, similar to other neighborhood selection methods, when estimating the neighborhood of a node, our method enjoys computational efficiency, as we only need to solve p individual regression problems. These neighborhood estimation procedures can be performed in parallel, leading to a highly efficient estimation process. Fourth, to the best of our knowledge, this is the first time that FGMs have been explored using a DNN framework. These advancements enable us to analyze this type of data in a much more flexible manner, which may not have been possible with other methods. Fifth, the proposed DNN framework effectively identified key connectivity differences in real-world neuroimaging data, particularly highlighting reduced cerebellar, precuneus, and frontal connectivity in ADHD compared to controls. This underscores the method's ability to reveal nuanced disruptions in regions critical for cognitive and attentional functions, offering deeper insights into ADHD-related neurobiology.

1.3. Related Work

This work is related to a series of papers on graphical models, including the *graphical Lasso* (Yuan and Lin 2006; Friedman, Hastie, and Tibshirani 2008) and the *neighborhood selection* approach (Meinshausen and Bühlmann 2006). Most existing literature on FGMs extends these methods, specifically the graphical Lasso and neighborhood selection, to the functional data setting. For example, Qiao, Guo, and James (2019) proposed the *functional graphical Lasso*, which induces sparse estimation of the precision matrix of the Gaussian likelihood through L_1 regularization. Various extensions, computational techniques, and theoretical properties have been developed within these frameworks. Additionally, Zhu, Strawn, and Dunson (2016) constructed a Bayesian framework for FGMs by establishing Markov distributions and hyper Markov laws for random processes. The *neighborhood selection* approach, originally introduced by Meinshausen and Bühlmann (2006), involves estimating the neighborhood of each node separately by regressing each variable on all the remaining variables, using L_1 regularization to induce sparsity, and then combining the neighborhoods to form the global graph estimate. This method was initially developed to estimate the conditional independence graph when the vector-valued data are drawn from a multivariate Gaussian distribution. More recently, Zhao et al. (2024) proposed a neighborhood selection approach specifically for Gaussian FGMs, in which they estimate the neighborhood of each node via a function-on-function regression and subsequently recover the entire graph structure by combining the estimated neighborhoods.

As aforementioned motivation, Gaussian process is the key assumption in most existing work on FGMs. For instance, Qiao, Guo, and James (2019); Zhu, Strawn, and Dunson (2016); Zhao et al. (2024) are developed within the framework of multivariate Gaussian processes. Similarly, Zapata, Oh, and Petersen (2021) exploited separability in multivariate functional data and introduced a partially separable Karhunen-Loève expansion for functional Gaussian graphical models. Some approaches have allowed for non-Gaussianity, such as the nonparanormal approach proposed by Solea and Li (2022), which uses copula functions to estimate a joint non-Gaussian density while preserving conditional independence. However, this approach is still limited by the choice of the copula function, and, to the best of our knowledge, no fully general approach is currently available. In another direction, Li and Solea (2018) replaced conditional independence with the concept of functional conditional additive independence to relax the Gaussian assumption. A similar idea was adopted in the recent work of Solea and Dette (2022).

As a more powerful tool to characterize the complex functions, neural network methods are recently used for functional data analysis. For example, Thind, Multani, and Cao (2023) proposed functional neural network taking functional inputs and covariates through multi-layer perceptrons with functional weighting based on basis expansion. Wang and Cao (2023) proposed a nonlinear prediction model for functional time series by applying neural networks as a powerful functional approximation tool. Yao, Mueller, and Wang (2021) introduced neural networks that employ a Basis layer and applied parsimonious dimension reduction to functional inputs. In Wang et al. (2023b) and Wang, Cao, and Shang (2023a), the authors proposed an optimal DNN-based classifier for both densely and sparsely observed multi-dimensional functional data, while Wang, Cao, and Huang (2025) introduced a feature selection technique for multi-dimensional functional covariates in classification tasks. For a more detailed discussion of the development of DNN method for functional data analysis, we refer to recent review papers such as Wang et al. (2024b) and Wang, Cao, and Huang (2024a).

The remainder of the article is organized as follows. Section 2 describes the methodology and proposes the nonparametric FGM. Section 4 implements the estimation procedure. Section 3 provides some theoretical support for the proposed algorithm. In Section 5 we conduct simulation studies to evaluate the finite sample properties of the proposed methodology, and in Section 6 we apply the new model to two real datasets. We conclude with some final remarks in Section 7, while all proofs of the theoretical results, some specific model examples and additional numerical results are deferred to the supplementary file.

2. Formulation of Functional Graphical Model

2.1. Preliminaries

The common undirected graphical models describe the conditional dependencies among the components of a p-dimensional random vector $\boldsymbol{X}$. Now, we extend the notation of p-dimensional random vector $\boldsymbol{X}$ to p-dimensional multivariate functional data $\boldsymbol{X}(t) = (X^1(t), \ldots, X^p(t))^\intercal$, $t \in [0,1]$, where each node represents a random function rather than a random scalar, it represents FGM. The FGM of $\boldsymbol{X}(t)$ is represented by an undirected graph $G = (\mathbb{V}, \mathbb{E})$, with $\mathbb{V}$ representing the set of vertices and $\mathbb{E}$ the set of edges. Let $\mathbb{V} = \{1, \ldots, p\}$, and $\mathbb{E}$ denote a subset of $\{(i,j) \in \mathbb{V} \times \mathbb{V} : i \neq j\}$, which satisfies $(i,j) \in \mathbb{E}$ if and only if $(j,i) \in \mathbb{E}$, because for an undirected graph (i,j) and (j,i) represent the same edge. The set of edges $\mathbb{E}$ is defined by the relation $(i,j) \notin \mathbb{E} \Leftrightarrow X^i(t) \perp\!\!\!\perp X^j(t) | \boldsymbol{X}^{-(i,j)}(t)$, where $\perp\!\!\!\perp$ denotes statistical independence between two random process and $\boldsymbol{X}^{-(i,j)}(t)$ represents $\boldsymbol{X}(t)$ with its ith and jth components removed. Suppose we observe n iid p-dimensional multivariate functional training samples $\{\boldsymbol{X}_k(t) : 1 \leq k \leq n\}$, and each $X_k^i : [0,1] \to \mathbb{R}$ is assumed to be in $L^2([0,1])$, s.t. $\mathrm{E}\langle X_k^j(t), X_k^j(t)\rangle < \infty$ and $\langle \cdot, \cdot \rangle$ denotes inner product of two functions over $[0,1]$.

To represent functional data, there are two typical ways, using either pre-fixed basis functions or data-driven methods to express the underlying processes $\{X_k^i(t) : 1 \leq i \leq p, 1 \leq k \leq n\}$. Note that data-driven bases generally have to be estimated from p univariate functional principal component analysis procedures, which is computationally prohibited when $p \gg n$, see the discussion in Xue, Yang, and Yao (2023). Hence, we adopt a common pre-fixed basis that is complete and orthonormal in $L_2([0,1])$ for all processes of interest. We assume the data admit $X_k^i(t) = \sum_{\ell=1}^{\infty} \xi_{k,\ell}^i \phi_\ell(t)$, and can be estimated by $X_k^i(t) \approx \sum_{\ell=1}^{\infty} \widehat{\xi}_{k,\ell}^i \widehat{\phi}_\ell(t)$, where $\widehat{\boldsymbol{\xi}}_k^i := (\widehat{\xi}_{k,1}^i, \widehat{\xi}_{k,2}^i, \ldots)$ is an estimator of $\boldsymbol{\xi}_k^i := (\xi_{k,1}^i, \xi_{k,2}^i, \ldots)$ and $\boldsymbol{\xi}_k^i$ are unobservable random coefficients of X_k^i with respect to the population basis ϕ_ℓ. It's noteworthy that while $\widehat{\phi}_\ell$ may differ from the true basis function ϕ_ℓ, this dissimilarity does not impact the numerical performance of the proposed selection procedure. This finding is also supported by numerical results for finite samples in Section 5. Hence, it is natural to estimate FGM based on $\widehat{\boldsymbol{\xi}}_{kj}$'s and we do not further pursue other complicated basis-seeking procedures.

Let $\boldsymbol{\xi}_{(m_i)}^i = (\xi_1^i, \ldots, \xi_{m_i}^i)^\intercal$ be the vector of the first m_i functional projection scores in the Karhunen-Loève expansion of each X^i, $i = 1, \ldots, p$. In the following, To simplify the notation, we drop the subscript (m_i) from $\boldsymbol{\xi}_{(m_i)}^i$, that is $\boldsymbol{\xi}^i \equiv \boldsymbol{\xi}_{(m_i)}^i$. We denote $\boldsymbol{m} = (m_j)_{j=1,\ldots,p}^\intercal$ as the truncation vector of functional projection scores, and $\boldsymbol{m}^{(-i)} = (m_j)_{j \neq i}^\intercal$ as the truncation vector excluding the ith node of interest.

Let $\|\cdot\|_1$ be the L_1 norm, $\|\cdot\|_2$ be the L_2 norm, and $\|\cdot\|_\infty$ be the infinity norm. We express our nonparametric FGM as $\mathrm{E}\left(\boldsymbol{\xi}^i | \boldsymbol{\xi}^{(-i)}\right) = \boldsymbol{f}^i\left(\boldsymbol{\xi}^{(-i)}\right)$, where $\boldsymbol{\xi}^{(-i)} = \left(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^{i-1}, \boldsymbol{\xi}^{i+1}, \ldots, \boldsymbol{\xi}^p\right)$, $\mathrm{E}\left\{\boldsymbol{f}^i(\boldsymbol{\xi}^{(-i)})\right\} = \boldsymbol{0}$, and $\boldsymbol{f}^i : \mathbb{R}^{\|\boldsymbol{m}^{(-i)}\|_1} \to \mathbb{R}^{m_i}$. The following definition introduces the structure of the function class $\mathcal{F}^i$, within which $\boldsymbol{f}^i$ is defined.

Definition 1. (Hölder smooth function) Given $\boldsymbol{m}^{(-i)}$ and a collection of parameters $J^i \in \mathbb{N}_+$, $\boldsymbol{r}^i = \left(r_1^i, \ldots, r_{J^i}^i, r_{J^i+1}^i\right)^\intercal \in \mathbb{N}_+^{J^i+1}$, $\boldsymbol{s}^i = \left(s_1^i, \ldots, s_{J^i}^i\right)^\intercal \in \mathbb{N}_+^{J_i}$, $\boldsymbol{\gamma}^i = \left(\gamma_1^i, \ldots, \gamma_{J^i}^i\right)^\intercal \in \mathbb{N}_+^{J_i}$, $\boldsymbol{a}^i = \left(a_1^i, \ldots, a_{J^i}^i\right)^\intercal \in \mathbb{N}_+^{J_i}$, $\boldsymbol{b}^i = \left(b_1^i, \ldots, b_{J^i}^i\right)^\intercal \in \mathbb{N}_+^{J_i}$, it follows that each element $f_l^i : \mathbb{R}^{\|\boldsymbol{m}^{(-i)}\|_1} \to \mathbb{R}$ of $\boldsymbol{f}^i \in \mathcal{F}^i$ satisfies the expression of $f_l^i(\boldsymbol{x}) = \boldsymbol{e}_{J^i} \circ \ldots \circ \boldsymbol{e}_2 \circ \boldsymbol{e}_1(\boldsymbol{x})$, $\forall \boldsymbol{x} \in \left[a_1^i, b_1^i\right]^{\|\boldsymbol{m}^{(-i)}\|_1}$, where $\boldsymbol{e}_u : \left[a_u^i, b_u^i\right]^{r_u^i} \to \left[a_{u+1}^i, b_{u+1}^i\right]^{r_{u+1}^i}$, and each element $e_{uv} : \left[a_u^i, b_u^i\right]^{s_u^i} \to \left[a_{u+1}^i, b_{u+1}^i\right]$ of $\boldsymbol{e}_u$ are γ_u-Hölder smooth function of s_u^i covariates. By convention, $r_{J^i+1}^i = m_i$.

Remark 1. This is the basic concept of a Hölder smooth function, with the formal definition provided in the supplementary material due to space constraints. The rich function class collection $\left\{\mathcal{F}^i\right\}_{i=1}^p$ encompasses all existing models from the literature. For example, the functional linear model in Zhao et al. (2024) and additive model in Solea and Dette (2022). In the Supplement file, we further discuss the connection between the proposed compositional $\mathcal{F}^i$ and several popular FGMs.

It is worth noting that the dimension of input $\|\boldsymbol{m}^{(-i)}\|_1$ and output m_i are both diverging with n. Definition 1 addresses the intrinsic dimension of $\boldsymbol{f}^i$ can be controlled by vector $\boldsymbol{s}^i$. We

impose the following assumption for output $\boldsymbol{\xi}^i$ to achieve fast convergence.

Assumption 1. There exist $\tau_1, \tau_2 > 0$, such that $\mathrm{E}\left\{\exp\left(\tau_1 \|\boldsymbol{\xi}^i\|_\infty\right)\right\} \le \tau_2$.

Remark 2. This moment assumption for $\boldsymbol{\xi}^i$ is weaker than the sub-Gaussian assumption which is commonly used for nonparametric regression using DNN (Bauer and Kohler 2019; Schmidt-Hieber 2020). Additionally, it is trivial that when $\|\boldsymbol{\xi}^i - \widehat{\boldsymbol{\xi}}^i\|_\infty < \epsilon$ for some small constant $\epsilon > 0$, there exists a constant $\tau_2'(\tau_1, \tau_2, \epsilon)$, such that $\mathrm{E}\left\{\exp\left(\tau_1 \|\widehat{\boldsymbol{\xi}}^i\|_\infty\right)\right\} \le \tau_2'(\tau_1, \tau_2, \epsilon)$. This equivalence ensures that we can construct the proposed algorithm directly on the feasible estimators $\widehat{\boldsymbol{\xi}}^i$.

2.2. Deep Neural Network Functional Graphical Model

In this section, inspired by the rich approximation power of DNN, we propose a novel neighborhood selection procedure, which can approach nonparametric FGM closely even for complicated $\boldsymbol{f}^i(\cdot)$. Let $\sigma(x) = (x)_+$ be the ReLU activation function. For ith node, let $L^i \in \mathbb{N}^+$ denote the number of hidden layers, $\boldsymbol{d}^i = (d_1^i, \ldots, d_{L+1}^i) \in \mathbb{N}^{L^i+1}$ denotes the node vector of the hidden layer. Let $\boldsymbol{B}_j^i$ be a $m_i \times m_j$ weight matrix for the residual layer. The input layer and output layer are fixed by $d_1^i = \|\boldsymbol{m}^{(-i)}\|_1$ and $d_{L^i+1}^i = m_i$, respectively. Define the class of residual feed-forward neural networks with $\|\boldsymbol{m}^{(-i)}\|_1$ inputs by

$$\mathcal{G}^i = \left\{ \boldsymbol{g} = \sum_{j \ne i}^{p} \left(\boldsymbol{B}_j^i\right)^\intercal \widehat{\boldsymbol{\xi}}^j + \boldsymbol{h}_{\boldsymbol{W}^i}\left(\widehat{\boldsymbol{\xi}}^{(-i)}\right) \right\}, \tag{2.1}$$

such that any generic $h_{\boldsymbol{W}^i}$ has an expression

$$\boldsymbol{h}_{\boldsymbol{W}^i}(\boldsymbol{x}) = \mathbf{W}_{L^i}^i \sigma\left(\mathbf{W}_{L^i-1}^i \sigma(\ldots \mathbf{W}_2^i \sigma(\mathbf{W}_1^i \boldsymbol{x}) \ldots\right), \quad \boldsymbol{x} \in \mathbb{R}^{\|\boldsymbol{m}^{(-i)}\|_1},$$

where $\mathbf{W}_l^i \in \mathbb{R}^{d_{l+1} \times d_l}$, for $l = 1, \ldots, L^i$, are the weight matrices.

2.3. Functional Graphical Model Structure Selection

In the following, we consider a regularization approach with constraints on weights to sparsify the edge connections. This architecture involves multiple layers of neurons with Lasso regularization applied to the weights connecting the neurons in each layer. During training, the Lasso penalty term is added to the loss function, promoting a sparse solution by shrinking some weights toward zero. This encourages the model to focus on the most informative features while discarding irrelevant ones. Specifically, for the ith node, we estimate the regressor $\boldsymbol{f}^i$ by optimizing the following problem:

$$\widehat{\boldsymbol{f}}^i = \underset{\boldsymbol{g} \in \mathcal{G}^i}{\arg\min}\, n^{-1} \sum_{k=1}^{n} \left\| \boldsymbol{\xi}_k^i - \boldsymbol{g}\left(\boldsymbol{\xi}^{(-i)}\right) \right\|_2^2 + \lambda^i \sum_{j \ne i}^{p} P_j\left(\boldsymbol{B}_j^i\right),$$
$$\text{s.t. } \|\mathbf{W}_1^{i(j)}\|_\infty \le M \|\boldsymbol{B}_j^i\|_F, \quad j \ne i, \tag{2.2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm; $P_j(\cdot) : \mathbb{R}^{m_i \times m_j} \to \mathbb{R}$ is the penalty function for regularization; the hierarchy coefficient M controls the relative strength of the linear and nonlinear components; the penalty coefficient λ controls the complexity of the fitted model; and $\mathbf{W}_1^{i(j)}$ is the sub-matrix of $\mathbf{W}_1^i$ connected to $\widehat{\boldsymbol{\xi}}^j$. The regularizers $P_j(\cdot)$ impose sparsity at the group level for entire $\boldsymbol{B}_j^i$, meaning that the entire group-coefficients of $\widehat{\boldsymbol{\xi}}^j$'s can be shrunk to zero. The constraints are designed to calibrate the level of non-linearity attributed to feature j. If the entire matrix $\boldsymbol{B}_j^i$ are forced to be zero, then all weights connected to the neural network that are related to $\widehat{\boldsymbol{\xi}}^j$ are effectively shrunk to be zero, leading to the exclusion of the jth feature from the model of the ith node. This idea extends the LassoNet algorithms proposed in Lemhadri et al. (2021) for iid high-dimensional data, where P_j are set to be the group-LASSO penalty with respect to the Frobenius norm. A graphical representation is demonstrated in Figure S.1 in the supplementary material.

The selection of the tuning parameters is important in practice. This is not only because the DNN structure significantly impacts the model performance, but also due to the key distinction between the functional and multivariate data—the inclusion of the truncation vector $\boldsymbol{m}$, which substantially influences both classification accuracy and feature selection efficiency. Validation is a popular approach and is integrated into LassoNet, making it an intuitive first choice when extending the LassoNet algorithm. We also adopt the high-dimensional BIC (Wang, Kim, and Li 2013) for accurate edge selection. Specifically, let $\boldsymbol{\eta} = \left(\boldsymbol{m}^{(-i)}, L^i, \boldsymbol{d}^i, \lambda^i\right)$ be the tuning parameter set for optimization problem (2.2). We choose tuning parameters $\boldsymbol{\eta}$ that minimizes the following BIC criterion for FGM feature selection with respect to the i-th node of interest: $\text{FGM-BIC}\left(\boldsymbol{\eta}^i\right) = \sum_k \left\| \boldsymbol{\xi}_k^i - \widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}_k^{(-i)}; \boldsymbol{\eta}^i\right) \right\|_2 + \|\widehat{\mathcal{N}}_i\left(\boldsymbol{\eta}^i\right)\|_0 \frac{\log p}{n}$, where $\widehat{\mathcal{N}}_i\left(\boldsymbol{\eta}^i\right) = \widehat{\mathcal{N}}_i := \left\{ j : \|\widehat{\boldsymbol{B}}_j^i\|_F \ne 0, j \ne i \right\}$ is the estimate of $\mathcal{N}_i := \left\{ j : \|\boldsymbol{B}_j^i\|_F \ne 0, j \ne i \right\}$ with respect to the choice of $\boldsymbol{\eta}$. We name the associated algorithm "`fgDNN`". The following Algorithm 1 describes this procedure, and Theorem 3.4 in Section 3 validates the algorithms. Finally, we stitch the neighborhoods together to form the global graph estimate. By applying this approach, we are able to effectively estimate the true edge set $\mathbb{E}$ by

$$\widehat{\mathbb{E}} = \left\{ (i,j) \in \mathbb{V} \times \mathbb{V} : i \ne j, \|\widehat{\boldsymbol{B}}_j^i\|_F \ne 0 \text{ and } \|\widehat{\boldsymbol{B}}_i^j\|_F \ne 0 \right\}.$$

2.4. Learning Conditional Independence

In the classical setting, one of the most appealing features of the Gaussian graphical model is its ability to represent conditional independence through the zero entries of the precision matrix, reducing the problem of estimating an undirected graph to the sparse estimation of the precision matrix. Without assuming that X is a Gaussian random function, we explore conditional independence by examining the independence between the projection scores, denoted as $\boldsymbol{\xi}$. Let $\boldsymbol{\xi}^i = (\xi_1^i, \ldots, \xi_{m_i}^i)^\intercal$ represent the first m_i functional projection scores. We characterize the pairwise conditional independence of $\boldsymbol{\xi}^i$ using mixed second-order partial derivatives, as described in the following lemma.

Algorithm 1 `fgDNN` algorithm for FGM

Inputs: Functional data $\{\boldsymbol{X}_k(t)\}_{k=1}^n$, basis functions $\{\widehat{\phi}_1, \widehat{\phi}_2, \ldots,\}$, epoch number b, hierarchy coefficient M, learning rate α, training index $\mathcal{I}_1$, validation index $\mathcal{I}_2$, tuning candidates: $\{\boldsymbol{\eta}_1^i, \ldots, \boldsymbol{\eta}_N^i\}$ with initial value λ_0 and path multiplier δ,
for $i = 1, \ldots, p$ **do**
  for $\ell = 1, \ldots, N$ **do**
    Apply Algorithm 2 on $\{\boldsymbol{X}_k(t)\}_{k\in\mathcal{I}_1}$ and obtain $\widehat{\mathcal{N}}_i(\boldsymbol{\eta}_\ell^i)$, $\widehat{\boldsymbol{B}}_j^i(\boldsymbol{\eta}_\ell^i)$ and $\{\widehat{\mathbf{W}}_l^i(\boldsymbol{\eta}_\ell^i)\}_l$
    Obtain the estimated $\widehat{\boldsymbol{f}}^i\left(\widehat{\boldsymbol{\xi}}^{(-i)}; \boldsymbol{\eta}_\ell^i\right)$, $i \in \mathcal{I}_2$
  end for
  Obtain $\boldsymbol{\eta}^{i*} = \underset{\boldsymbol{\eta}\in\{\boldsymbol{\eta}_1^i,\ldots,\boldsymbol{\eta}_N^i\}}{\arg\min}$ FGM-BIC $(\boldsymbol{\eta}^i)$
  Obtain $\widehat{\boldsymbol{B}}_j^i(\boldsymbol{\eta}^{i*})$, $\{\widehat{\mathbf{W}}^i(\boldsymbol{\eta}^{i*})\}_l$, and $\widehat{\mathcal{N}}_i(\boldsymbol{\eta}^{i*})$
end for
Combine all neighborhoods into the estimated edge set $\widehat{\mathbb{E}}$
Outputs: Return $\widehat{\mathbb{E}}$

Algorithm 2 Training penalty term in Model (2.2)

Inputs: The jth functional data in the training set $\left\{X_k^j\right\}_{k\in\mathcal{I}_1}$, truncation vectors $\boldsymbol{m_i}$, basis functions $\left\{\widehat{\phi}_1, \ldots, \widehat{\phi}_{m_j}\right\}$, number of hidden layer L, number of neurons $\boldsymbol{d}$, epoch number b, hierarchy coefficient M, path multiplier δ, learning rate α, initial tuning parameter λ_0.
Initialize $\lambda = \lambda_0$, $p_0 = p$
Obtain $\widehat{\xi}_{k,\ell}^j = \langle X_k^j, \widehat{\phi}_\ell\rangle$, $k \in \mathcal{I}_1$, $\ell = 1, \ldots, m_j$
while $p_0 \geq 1$ **do** $\lambda \leftarrow (1+\delta)\lambda$
  for $u = 1, 2, \ldots, b$ **do**
    Compute $\nabla_{\boldsymbol{B}}\mathcal{L}$, $\nabla_{\boldsymbol{W}}\mathcal{L}$ by backpropagation, where $\mathcal{L}$ is the objective function in (2.2)
    Update $\boldsymbol{B} \leftarrow \boldsymbol{B} - \alpha\nabla_{\boldsymbol{B}}\mathcal{L}$, $\boldsymbol{W} \leftarrow \boldsymbol{W} - \alpha\nabla_{\boldsymbol{W}}\mathcal{L}$
    for $j = 1, \ldots, i-1, i+1, \ldots, p$ **do**
      Update $\widehat{\boldsymbol{B}}_j^i$ and $\widehat{\boldsymbol{W}}_1^{(j)}$ via Algorithm 3 in the supplement
    end for
  end for
  Update $\widehat{\mathcal{N}}^i$ to be the set of nonzero coordinates of $\|\widehat{\boldsymbol{B}}_j^i\|_F$
end while
Outputs: $\widehat{\mathcal{N}}^i$, $\widehat{\boldsymbol{B}}_j^i$, and $\{\widehat{\boldsymbol{W}}_l\}_l$

Lemma 2.1. (Conditional independence equivalence) If we assume that the joint density function $\pi(\cdot)$ is continuously differentiable, this yields that, for any $(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^p)$

$$\partial_{\boldsymbol{\xi}^i}\partial_{\boldsymbol{\xi}^j}\log\pi(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^p) = \mathbf{0} \Leftrightarrow \boldsymbol{\xi}^i \perp\!\!\!\perp \boldsymbol{\xi}^j | \boldsymbol{\xi}^{-(i,j)}.$$

Given two functional data $X^i(t)$ and $X^j(t)$ for $i \neq j$, let $X^i(t) \perp\!\!\!\perp X^j(t)|\boldsymbol{X}^{-(i,j)}$ denote that the lack of an edge between nodes i and j and the two functional data are conditionally independent given the remaining variables. In most cases, the sequence of eigenvalues decreases fast. Hence, without much loss of generality, we assume that the first m_i functional projection scores, $\boldsymbol{\xi}^i$, sufficiently represent the distribution of the functional data $X^i(t)$, where m_i is sufficiently large and converges to infinity as the sample size grows. Hence, when m_i and m_j are large enough, $\boldsymbol{\xi}^i \perp\!\!\!\perp \boldsymbol{\xi}^j|\boldsymbol{\xi}^{-(i,j)} \Leftrightarrow X^i(t) \perp\!\!\!\perp X^j(t)|\boldsymbol{X}^{-(i,j)}$ and $\partial_{\boldsymbol{\xi}^i}\partial_{\boldsymbol{\xi}^j}\log\pi(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^p) = \mathbf{0} \Leftrightarrow X^i(t) \perp\!\!\!\perp X^j(t)|\boldsymbol{X}^{-(i,j)}$.

Given the neighborhood learning in Section 2.2, for any $i = 1, \ldots, p$, we have $\boldsymbol{\xi}^i = \widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}^{(-i)}\right)$. If $\boldsymbol{\xi}^i = \widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}^{-(i,j)}\right)$, we have

$$\begin{aligned}\pi(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^p) &= \pi(\boldsymbol{\xi}^i, \boldsymbol{\xi}^j, \boldsymbol{\xi}^{-(i,j)})\\ &= \pi\left(\widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}^{-(i,j)}\right), \widehat{\boldsymbol{f}}^j\left(\boldsymbol{\xi}^{-(i,j)}\right), \boldsymbol{\xi}^{-(i,j)}\right)\\ &= \pi\left(\widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}^{-(i,j)}\right)\right)\pi\left(\widehat{\boldsymbol{f}}^j\left(\boldsymbol{\xi}^{-(i,j)}\right)\right)\pi\left(\boldsymbol{\xi}^{-(i,j)}\right)\end{aligned}$$

which easily leads to Equation (S.2.1) in the supplement. Observe that the condition $\partial_{\boldsymbol{\xi}^i}\partial_{\boldsymbol{\xi}^j}\log\pi(\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^p) = \mathbf{0}$ is *necessary, but generally not sufficient*, for the working regression specification $\boldsymbol{\xi}^i = \widehat{\boldsymbol{f}}^i(\boldsymbol{\xi}^{-(i,j)})$. Local regressions based solely on this criterion can still register weak or spurious associations, leading to an overly dense neighborhood set. To curtail such false positives, we thereby adopt a symmetric selection rule (Section 2.3) that declares an undirected edge only when *both* directed selections concur. This "and" requirement retains truly adjacent nodes while excising extraneous links, thereby aligning the estimated graph with the conditional-independence structure articulated in Lemma 2.1.

3. Theoretical Properties

We now discuss the statistical properties of the estimator proposed in Section 2.3. The following assumption describes the optimal DNN structure of $\mathcal{G}^i$ to approximate $\mathcal{F}^i$. For two real positive sequences $\{a_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$, we write $a_n \lesssim b_n$ when there exists universal constants c and $n_0 \in \mathbb{N}^+$, such that $a_n/b_n \leq c$ for any $n \geq n_0$.

Assumption 2. For any node $i \in \{1, \ldots, p\}$, the DNN class $\mathcal{G}^i$ satisfies that

(a) $L^i \geq J^i + 216\sum_{u=1}^{J^i}\left(\lceil\gamma_u^i\rceil^2 + 1\right)$;

(b) $\min_{2\leq v\leq L^i} d_v^i \geq 81 m_i \max_{1\leq u\leq J^i} r_u^i\left(\lceil\gamma_u^i\rceil + s_u^i + 2\right)^{s_u^i+1} 3^{s_u^i+1}$;

(c) $\|\boldsymbol{m}^{(-i)}\|_1 \lesssim L^i \max_{2\leq v\leq L^i} d_v^i \asymp L^i \min_{2\leq v\leq L^i} d_v^i \asymp n^{\frac{\tilde{s}^i}{2(2\tilde{\gamma}^i+\tilde{s}^i)}}$, with $\tilde{\gamma}^i = \tilde{\gamma}_{\tilde{u}^i}^i$, $\tilde{s}^i = s_{\tilde{u}^i}^i$, where $\tilde{u}^i = \underset{1\leq u\leq J^i}{\arg\min}\frac{\tilde{\gamma}^i}{s_u^i}$, and $\tilde{\gamma}_u^i = \gamma_u^i\prod_{v=u+1}^{J^i}\left(\gamma_v^i \wedge 1\right)$;

(d) $m_i \lesssim \log n$.

Remark 3. The critical aspect of Assumption 2 lies in the dimensions of $\boldsymbol{m}^{(-i)}$ and m^i, as it is both theoretically and practically infeasible to handle unbounded growth in $\boldsymbol{m}^{(-i)}$ and m^i for the regression task. We impose a relatively mild assumption on the truncation of $\widehat{\boldsymbol{\xi}}^i$, constrained by the order of $\log n$. Additionally, to ensure the fast convergence rate of $\boldsymbol{f}^i$, this condition also guarantees that the estimation of $\boldsymbol{\xi}^i$ captures sufficient information

to recover $X_i(t)$ as n approaches infinity, which is statistically efficient according to well-established literature.

In the following, we present the rate of convergence of our proposed estimator $\widehat{\boldsymbol{f}}^i$ given $\mathcal{N}^i$.

Theorem 3.1. (Consistency for the ith node) Under Assumptions 1 and 2, for any node $i \in \{1, \ldots, p\}$, one has

$$\sup_{\boldsymbol{f}^i \in \mathcal{F}^i} \|\widehat{\boldsymbol{f}}^i - \boldsymbol{f}^i\|_{n,\infty} = O_p\left(n^{-\frac{\tilde{\gamma}^i}{2\tilde{\gamma}^i + \tilde{s}^i}} \log^9(n)\right).$$

When implementing the proposed `fgDNN` across p nodes, a natural question is whether a universal DNN structure can be applied to all models. The following model demonstrates that such a structure exists and ensures fast convergence.

Theorem 3.2. (Uniform consistency) Under Assumptions 1 and 3 in the supplement, it follows that

$$\sup_{i=1,\ldots,p} \sup_{\boldsymbol{f}^i \in \mathcal{F}^i} \|\widehat{\boldsymbol{f}}^i - \boldsymbol{f}^i\|_{n,\infty} = O_p\left(n^{-\frac{\tilde{\gamma}}{2\tilde{\gamma} + \tilde{s}}} \log^9(n)\right).$$

Remark 4. Theorems 3.1 and 3.2 show that when the functional score for the ith node is controlled under the order of $\log n$, with properly chosen DNN class $\mathcal{G}^i$ ($\mathcal{G}$), the convergence rate reaches to the classical nonparametric regression rate (Stone 1982) up to a logarithmic factor, which is free of the extrinsic dimension $\boldsymbol{m}^{(-i)}$ and p. Theorem 3.2 also shows that a unified DNN structure exists for performing neighborhood selection across all nodes, ensuring fast convergence rates, particularly when $\boldsymbol{f}^i$ does not vary significantly.

Practically, the ith functional covariate trajectory is observed on N_i discrete locations. The following theorem provides the phase transition rate for the ih node, where the remaining functional covariates are observed on $\{N_1, \ldots, N_{i-1}, N_{i+1}, \ldots, N_p\}$ locations at a certain rate with respect to υ.

Theorem 3.3. (Consistency with sampling frequency) Under Assumptions 1 and 4 in the supplement, for some $\upsilon > 0$, if $\|\boldsymbol{\xi}^j - \widehat{\boldsymbol{\xi}}^j\|_\infty = O_p\left(N_j^{-\upsilon}\right)$ for any $j \neq i$, it follows that $\sup_{\boldsymbol{f}^i \in \mathcal{F}^i} \|\widehat{\boldsymbol{f}}^i\left(\widehat{\boldsymbol{\xi}}^{(-i)}\right) - \boldsymbol{f}^i\left(\boldsymbol{\xi}^{-i}\right)\|_{n,\infty} = O_p\left(n^{-\frac{\tilde{\gamma}^i}{2\tilde{\gamma}^i + \tilde{s}^i}} \log^9(n)\right) \mathbb{I}(\underline{N}_i \geq N_i^*) + O_p\left(\underline{N}_i^{-\upsilon}\right) \mathbb{I}(\underline{N}_i < N_i^*)$.

Remark 5. Theorem 3.3 demonstrates that, for neighborhood detection of the ith node, N_i^* serves as a critical sampling frequency across all functional covariates, governing the convergence rate over the function space $\mathcal{F}^i$ at any fixed sample size n. When $\underline{N}_i \geq N_i^*$, convergence rate is free from $\{N_j\}_{j \neq i}$ and consistent with the rate derived in Theorem 3.1. When $\underline{N}_i < N_i^*$, the convergence rate only depends on the sampling frequency of the most sparsely observed functional covariates. Assumption 4 introduces an adaptive DNN structure based on the N_i^*, ensuring a fast convergence rate of $\boldsymbol{f}^i$.

The following theorem guarantees that the solution $\widehat{\mathcal{N}}^i$ to Algorithm 2 is consistent with the true neighborhood set.

Theorem 3.4. (Global optimality) For any $0 \leq m_j < \infty$, $j = 1, \ldots, p$, the solution to Algorithm 2 is the global optimum of the constraint objective function (2.2).

4. Implementation

Our proposed neural networks are implemented using `Python` within the `PyTorch` framework. Specifically, the feedforward neural network is constructed based upon the number of hidden layer L, number of neurons in each hidden layer p_l, $l = 1, \ldots, L$. Throughout, the epoch number and batch size for training the neural network are fixed before the training process. Practically, we choose the epoch number to be moderately large, as long as the results are stable, and the batch size equals $2^{\lfloor \log n \rfloor}$, where $\lfloor \cdot \rfloor$ denotes the floor function. We implement Algorithm 1 to simultaneously determine the neural-network architecture and its truncation vectors. The procedure is calibrated to Theorem 3.1, ensuring that the resulting estimator attains the optimal convergence rate. Accordingly, candidate architectures are selected based on the asymptotic orders prescribed in Assumption 4 in the supplementary material, thereby keeping the empirical search aligned with the theoretically optimal design. For all numerical studies, we apply Algorithm 1 by randomly assigning 80% data to the training set and 20% data to the validation set. To optimize the training process with respect to objective (2.2), we use `Adam` optimizer with a learning rate of $\alpha = 0.001$ to train the initial model. The path multiplier is stable across all simulation settings, where we let $\delta = 0.02$. We choose hierarchy coefficient $M = 100$ as default. The optimization of Algorithm 2 with respect to the tuning parameter λ adopts the dense-to-sparse warm start approach. This approach yields effective performance with strong generalization ability (Lemhadri et al. 2021). We adopted the B-spline basis with order 4 to extract the functional principal scores. In practice, comparable performance can be obtained with other reasonable choices of basis functions. Python codes and examples for the proposed `fgDNN` algorithms are available on `GitHub` (*https://github.com/ShuoyangWang/fgDNN*).

5. Simulation

We illustrate the finite sample properties of our neighborhood selection procedure through a simulation study. For each setting, we fix sample size $n = 100$. Each random function is observed on grid with $T = 100$ equally spaced points on $[0, 1]$. We consider different dimension of $\mathbb{E}$ with $p = 30, 50, 100$. We generate data using the following four settings, which include the Gaussian, Copula Gaussian, and two nonparametric settings. In terms of the basis function $\widehat{\phi}_\ell$ selection, we use a common pre-fixed basis function, B-spline basis function in our proposed method. As discussed in Section 4, we perform a hyperparameter grid search over the number of hidden layers $\{1, 2, 3\}$, neurons per layer $\{30, 50, 100\}$, dropout rates $\{0.1, 0.3, 0.5\}$, and number of scores $\{5, 7, 9\}$.

Model A (Gaussian) Let $X_{ij}(t) = \boldsymbol{a}_{ij}\boldsymbol{f}(t)$, $i = 1, \ldots, n$, $j = 1, \ldots, p$, where $\boldsymbol{a}_{ij} \in R^{r^*}$ follows a mean zero Gaussian distribution with covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{\Theta}^{-1}$, and $\boldsymbol{f}(t)$ is a vector that contains the first r^* Fourier basis functions. We

generate a block-banded precision matrix $\boldsymbol{\Theta} \in R^{pr^* \times pr^*}$ with $r^* = 15$. Let $\boldsymbol{A} \in R^{r^* \times r^*}$ be a tridiagonal matrix with $\boldsymbol{A}_{kk} = 1$ and $\boldsymbol{A}_{k,k+1} = \boldsymbol{A}_{k+1,k} = 0.5$. The blocks of the precision matrix $\boldsymbol{\Theta}$ are then given as $\boldsymbol{\Theta}_{jj} = \boldsymbol{A}$, $\boldsymbol{\Theta}_{j,j+1} = \boldsymbol{\Theta}_{j+1,j} = \mathbf{0}.4A$, and $\boldsymbol{\Theta}_{j,j+2} = \boldsymbol{\Theta}_{j+2,j} = \mathbf{0}.2A$. All remaining blocks of $\boldsymbol{\Theta}$ are set to 0. This model has been studied in Zhao et al. (2024).

Model B (Copula Gaussian) To simulate copula Gaussian random functions, we first draw n independent Gaussian random functions using Model A. We then transform ξ^i_{ur} to $c_r(\xi^i_{ur})$, which are taken to be the following functions: $c_1(x) = x^3$, $c_2(x) = \exp(x)$, $c_3(x) = \exp(x)/(1 + \exp(x))$, $c_4(x) = (1+x)^5$, $c_5(x) = x$. This data generating setting exactly aligns with the model assumption in Solea and Li (2022).

Model C (Sparse nonparametric) Given a directed acyclic graph (DAG), $\mathbb{G} = (\mathbb{V}, \mathbb{E})$, we generate non-Gaussian scores by following the structural equations, $\xi^i_q = \sum_{j \in pa(i)} \sum_{r=1}^5 f^{ij}_{qr}(\xi^j_r) + \tilde{\epsilon}^j_q$, $i = 1, \ldots, p$, $q = 1, \ldots, 5$, where $pa(i)$ is the set of parents of node i in the associated DAG, $\tilde{\epsilon}^j_q \sim N(0, 1)$, and f^{ij}_{qr} are nonlinear and smooth functions and $f(x) = 1.4 + 3x - 0.5 + \sin(2\pi(x - 0.5)) + 8(x - 1/3)^2 - 8/9$. We simulate ξ^j_r independently from the uniform distribution $[0.5, 0.5]$, for all $r = 1, \ldots, 5$, $u = 1, \ldots, n$ and $i \in \mathbb{V}$. Next, we moralized the directed graph by simply removing the arrows the DAG to obtain the corresponding undirected graphs, which are used as the targets of estimation. Finally, we generate the multivariate functional data as $X^i_u(t_s) = \sum_{j \in pa(i)} \sum_{r=1}^5 \sum_{q=1}^5 f^{ij}_{qr}(\xi^j_r)\phi_q(t_s) + \epsilon^i_{us}$, $u = 1, \ldots, n$, where ϕ^i_q are the first five functions of the orthonormal Fourier basis, and ϵ^i_{us} is an iid sample from $N(0, 0.5^2)$ distribution. This data generating setting is analogue to the case in Solea and Dette (2022). We consider directed acyclic graphs with p nodes and three pairs of vertices are randomly selected as edges.

Model D (Dense nonparametric) The data is generated under the same setting in Model C, but 15 pairs of vertices are randomly selected as edges.

We compare our method with the classical functional Graphical lasso (FGLasso) procedure proposed by Qiao, Guo, and James (2019), the partial separability Karhunen-Loève expansion (PSKL) procedure proposed by Zapata, Oh, and Petersen (2021), the neighborhood selection procedure as a function-on-function regression procedure proposed by Zhao et al. (2024), nonparametric FGM (npFGM) procedure proposed by Solea and Dette (2022). For FGLasso, we select the parameters as proposed in Qiao, Guo, and James (2019). For PSKL, we use the package "fgm" with the default setting in Zapata, Oh, and Petersen (2021). Following Zhao et al. (2024), we implemented their methods in two different procedures on estimating FPCA. The first method, which we call g_X, projects each function onto its own FPCA basis and uses those projection scores for all subsequent tasks. The second method, which we call g_Y, projects all other functions onto the FPCA basis of X^i when selecting neighborhood $\mathcal{N}^i$. Following Zhao et al. (2024), the edge set can be computed by one of the following schemes: AND (if both $i \in \widehat{\mathcal{N}}_j$ and $j \in \widehat{\mathcal{N}}_i$ hold, then $(i, j) \in \widehat{\mathbb{E}}$) and OR (if either $i \in \widehat{\mathcal{N}}_j$ or $j \in \widehat{\mathcal{N}}_i$ holds, then $(i, j) \in \widehat{\mathbb{E}}$). Hence, we implement their methods as four different procedures: AND-g_X, AND-g_Y, OR-g_X, and OR-g_Y. The npFGM procedure is implemented by the source code provided by the authors of Solea and Dette (2022).

To compare the methods, we plot the respective Receiver Operating Characteristic (ROC) curve for each model and different values of p over 50 replications. For each value of the tuning parameter λ, we compare the estimated edge set to the true edge set. Specifically, we calculate the true positive rate $TPR(\lambda) = TP(\lambda)/(TP(\lambda) + FN(\lambda))$ and the false positive rate $FPR(\lambda) = FP(\lambda)/(FP(\lambda)+TN(\lambda))$, where $TP(\lambda)$, $FP(\lambda)$, $TN(\lambda)$, and $FN(\lambda)$ stand for the number of true positive, false positive, true negative, and false negative number of edges, respectively. The ROC curves are plotted by varying the penalty parameter λ, with TPR(λ) on the vertical axis and FPR(λ) on the horizontal axis. We also calculate the area under the ROC curve (AUC). The ROC curves are shown in Figures 1 and 2. The average AUC and its standard error are given in Table 1.

In Model A, where the functional data follows a Gaussian process, the PSKL procedure slightly outperforms `fgDNN` and the other methods when the number of nodes is relatively small. This is expected, as PSKL is specifically designed for functional Gaussian graphical models. We also observe that in Model A, the neighborhood selection procedure using the FPCA basis and the "optimal" PSKL basis performs comparably to `fgDNN`, which does not rely on prior knowledge of the basis function but instead uses a pre-selected basis. This result aligns with the efficient projection scores method discussed in Section 2.1.

However, when the multivariate Gaussian process assumption does not hold, our proposed `fgDNN` significantly outperforms the other methods in Models B, C, and D, across different numbers of nodes. Notably, as the conditional distribution becomes more complex-for instance, when 15 out of $p = 100$ nodes have edges-the PSKL method, which performs well in the Gaussian setting, becomes ineffective. Additionally, the npFGM method, which is designed for non-Gaussian settings, also underperforms compared to our proposed method in Models B, C, and D. We also present the numerical comparison for larger sample sizes ($n = 200, 300$) in Table 2 in the supplementary file. Our proposed method continues to outperform across all four models.

6. Applications: Attention Deficit Hyperactivity Disorder Data

Since the year of 2011, functional neuroimaging investigators working on Attention-Deficit Hyperactivity Disorder (ADHD) in three continents came together to form the ADHD-200 Consortium. The effort was to establish a large-scale, aggregate resting state fMRI dataset, along with accompanying anatomical and phenotypic data for children and adolescents with ADHD. We apply our proposed procedure to data from ADHD-200 Consortium Consortium (2012). The samples contain whole-brain fMRI scans for 74 ADHD patients that are diagnosed with ADHD, as well as 109 controls. We analyze the preprocessed fMRI dataset of Zhao et al. (2024), in which the whole-brain volumes are divided into $p = 116$ regions of interest (ROIs) and recorded at 232 time points per ROI. The researchers found that children with ADHD exhibited disrupted functional brain connectivity compared to typically

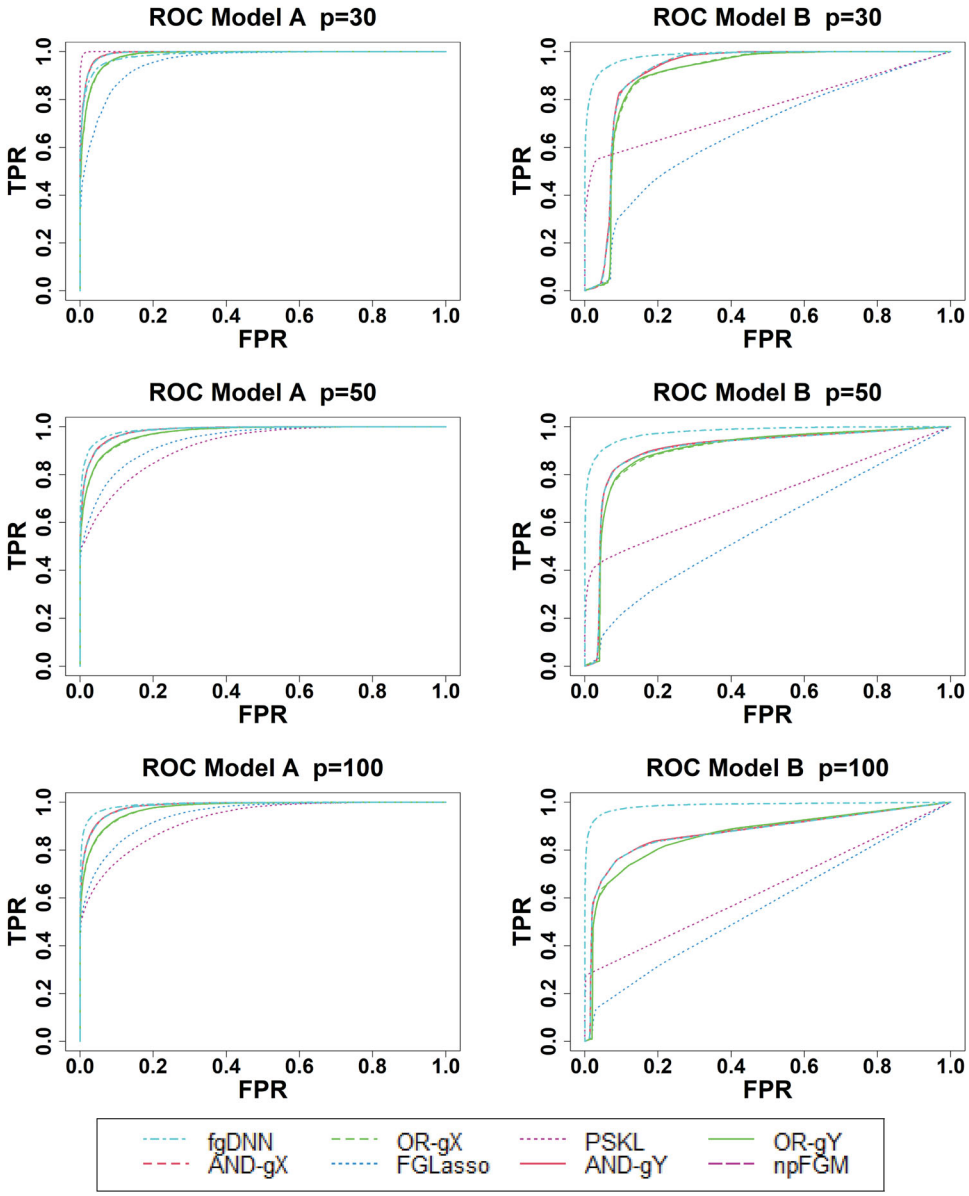

Figure 1. ROC curves for Models A and B.

Table 1. The average AUC for each method across 50 replicates; the standard errors of AUC are given in the parentheses.

Model	p	fgDNN	FPCA-g_Y,AND	FPCA-g_Y,OR	FPCA-g_X,AND	FPCA-g_X,OR	FGLasso	PSKL	npFGM
	30	0.993(0.003)	0.993(0.003)	0.984(0.005)	0.993(0.003)	0.985(0.004)	0.956(0.010)	**0.999(0.001)**	0.931(0.012)
	50	**0.992(0.003)**	0.984(0.004)	0.974(0.007)	0.984(0.005)	0.974(0.007)	0.942(0.010)	0.920(0.010)	0.942(0.008)
A	100	**0.991(0.002)**	0.985(0.003)	0.976(0.004)	0.984(0.003)	0.975(0.004)	0.947(0.006)	0.925(0.007)	0.952(0.005)
	30	**0.985(0.005)**	0.912(0.016)	0.895(0.021)	0.912(0.015)	0.895(0.022)	0.666(0.148)	0.766(0.193)	0.894(0.012)
	50	**0.986(0.006)**	0.903(0.009)	0.895(0.079)	0.905(0.088)	0.898(0.076)	0.576(0.149)	0.709(0.207)	0.917(0.008)
B	100	**0.987(0.003)**	0.873(0.140)	0.863(0.128)	0.875(0.145)	0.864(0.129)	0.566(0.131)	0.637(0.138)	0.937(0.004)
	30	**0.954(0.016)**	0.509(0.132)	0.538(0.151)	0.527(0.115)	0.576(0.123)	0.528(0.154)	0.561(0.099)	0.514(0.117)
	50	**0.973(0.008)**	0.517(0.170)	0.515(0.168)	0.498(0.124)	0.470(0.133)	0.516(0.160)	0.570(0.147)	0.544(0.145)
C	100	**0.986(0.006)**	0.541(0.170)	0.545(0.170)	0.546(0.166)	0.542(0.160)	0.536(0.160)	0.558(0.144)	0.507(0.119)
	30	**0.831(0.027)**	0.500(0.063)	0.500(0.068)	0.507(0.070)	0.504(0.069)	0.493(0.054)	0.517(0.001)	0.495(0.053)
	50	**0.885(0.016)**	0.495(0.056)	0.479(0.066)	0.501(0.043)	0.486(0.057)	0.491(0.054)	0.510(0.027)	0.487(0.051)
D	100	**0.934(0.009)**	0.497(0.045)	0.504(0.060)	0.504(0.049)	0.504(0.066)	0.500(0.061)	N/A(N/A)	0.504(0.059)

NOTE: The best result of each row is marked in bold text.

developing children (Wang et al. 2020). They observed altered connectivity patterns in several brain regions, including those associated with attention and cognitive control. Our goal is to recover the functional conditional independence graph for both groups and verify the aforementioned scientific findings.

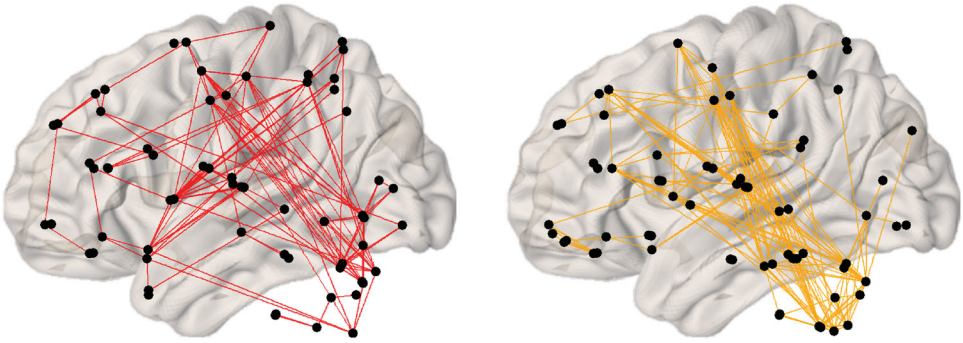

Figure 4. 3D brain connectome graph of ADHD (left panel) and control groups (right panel) by `fgDNN`.

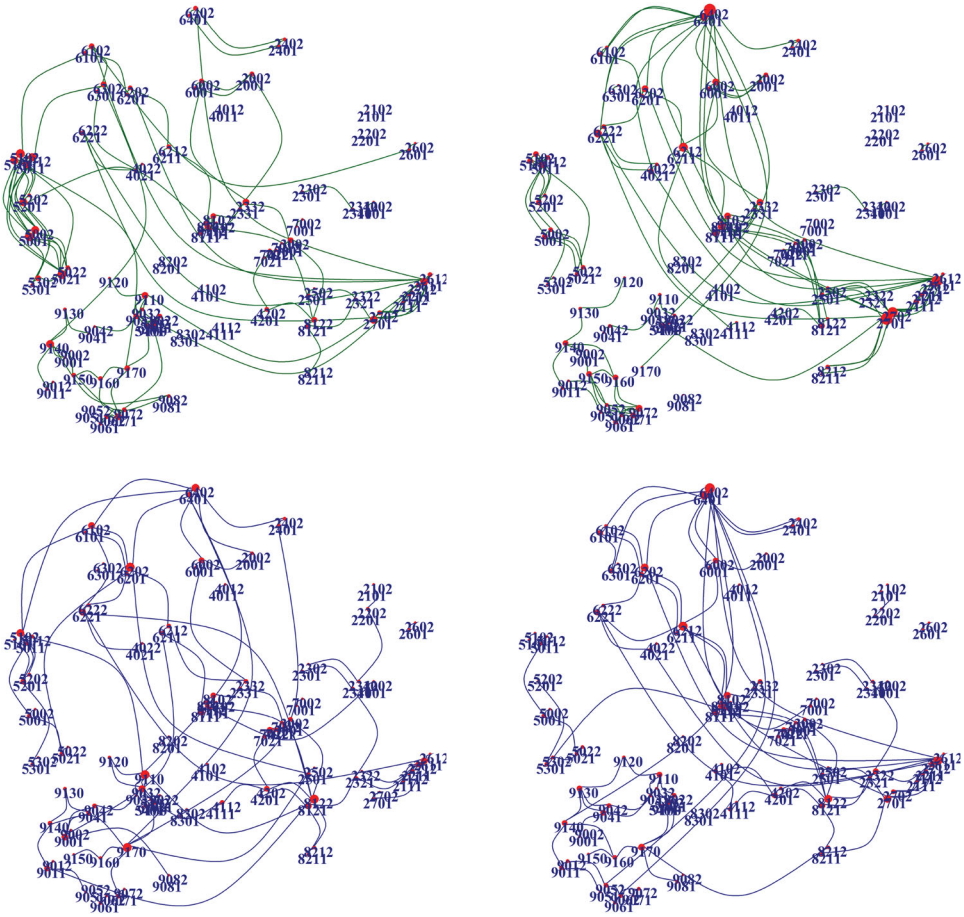

Figure 5. 2D Brain connectome graph of ADHD (the first column) and control (the second column) groups by AND-g_X (the first row) and npFGM (the second row).

We initially estimate the conditional independence graphs for both the ADHD and control groups using the popular BIC procedure with our proposed `fgDNN` method. In Figures 3 and 4, we present the estimated 2D and 3D brain connectome graph. The connectivity graph of the ADHD group contains 153 edges, while the control group has 173 edges, indicating a significant reduction in brain connectivity in the ADHD group across the entire brain network. In addition, the spatially dispersed connectivity in ADHD contrasts with the more concentrated patterns in controls, potentially reflecting compensatory or inefficient network adaptations. Specifically, connections in the cerebellum regions (nodes 9001, 9052, 9061, 9072), the precuneus regions (nodes 6101, 6102, 6302, and 6301 at the top left of the graph), and the frontal regions (nodes 2111 and 2112 on the right of the graph) are notably reduced in the ADHD patients. This diminished activity in ADHD has also been identified in the neuroscience literature (Wang et al. 2020). Furthermore, a marked difference between the ADHD and control groups can be observed in the cerebellum regions (nodes 9001 and 9002 at the bottom left of the graph). The cerebellum, traditionally known for its role in motor control, is now recognized as playing a key role in cognitive functions, including attention, working memory, and executive control. In ADHD, altered cerebellar connectivity may contribute to deficits in these areas (Wolfers et al. 2016).

We also applied the npFGM and AND-g_X methods to the ADHD dataset. To ensure a fair comparison, we adjusted the connection sparsity to 2% by tuning the penalty parameters for

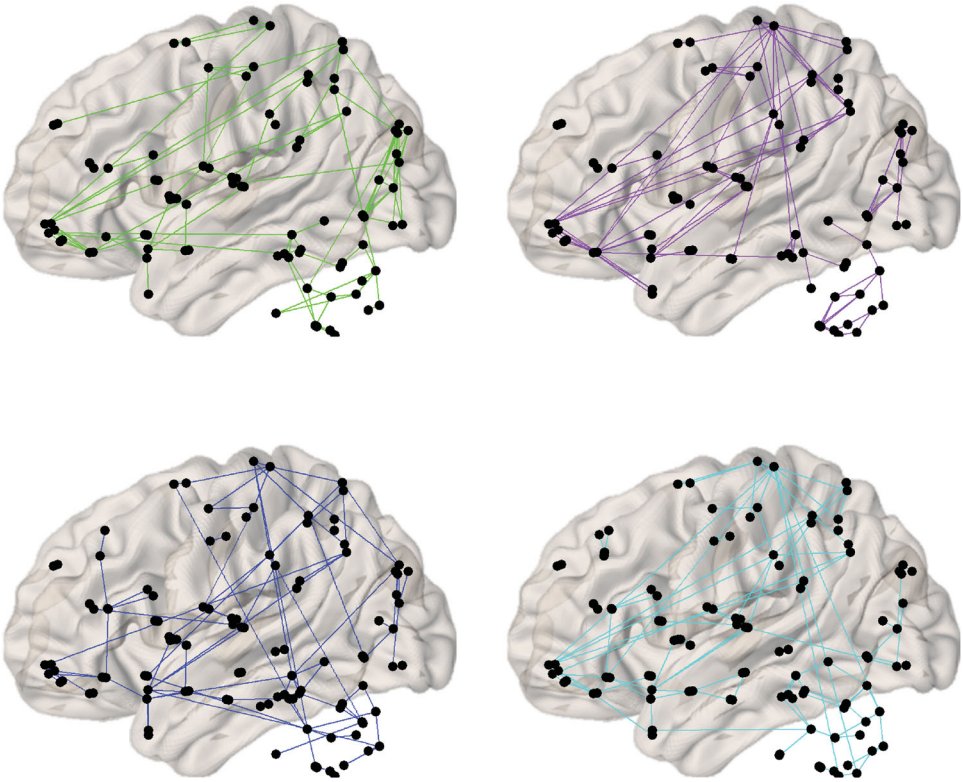

Figure 6. 3D Brain connectome graph of ADHD (the first column) and control groups (the second column) by npFGM (the first row) and AND-g_X (the second row) with the sparsity fixed to 2%.

both methods. For npFGM, we followed the recommendations of Solea and Dette (2022), using $(4 + \sqrt{n})$ B-spline functions of order 4, with the number of scores set to 5, and a tuning parameter selected to include 2% of the pairs. As suggested by Zhao et al. (2024), since all AND and OR methods yield comparable results, we implemented only the AND-g_X algorithm for illustrative purposes, following the procedures outlined in Zhao et al. (2024). It is followed the procedures outlined in Zhao et al. (2024). The resulting 2D graphs are shown in Figure 5, and the corresponding 3D graphs are shown in Figure 6. Since the `fgDNN` results in Figure 3 have sparsities of 2.3% and 2.6%, this allows for a relatively fair comparison among the three methods. Compared to the `fgDNN` method, both alternative methods display weaker signals. This suggests that `fgDNN` is more effective in identifying the most active regions in the brain connectivity graphs.

7. Summary

We introduce a novel DNN-based neighborhood selection procedure for multivariate functional data and FGMs. By leveraging the powerful DNN framework along with a new feature selection procedure, our method eliminates the need for the marginal Gaussian assumption while simultaneously introducing a fully nonparametric dependence structure, providing greater flexibility. This approach is particularly advantageous for handling large and complex multivariate functional data and FGMs. Theoretical guidance is provided to support the proposed algorithm. Our results demonstrate a marked reduction in brain connectivity in ADHD patients compared to controls, consistent with prior studies on disrupted connectivity in ADHD.

Supplementary Materials

Supplementary Materials: The supplemental files include the details for Algorithm 3, specific model examples showing that the other models are encompassed in our function class, the proofs of technical lemmas and Theorems 3.1–3.4, additional simulation results for phase transition phenomenon, and summary of notations used in Section 2. (FGDNNsupp.pdf)

Python code: The supplemental files for this article include the Python code needed to replicate the simulation study and real data analysis presented in the article. See the README contained in the zip file for more details. (fgDNNcode.zip)

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